



Supplementary Materials for

Generative AI use and misuse call for assessment reform in higher education

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1 Data and Measures

1.1 Data

This study uses data from the Student Experience in the Research University (SERU) Consortium, based at the Center for Studies in Higher Education at UC Berkeley. SERU is a comprehensive census-based survey conducted between March and August 2024 across 20 major public research-intensive universities in the United States. These institutions represent research-intensive (R1 by Carnegie Classification) universities in the U.S. that collectively enroll over 2.6 million undergraduates and award more than half of all bachelor's degrees in science, technology, engineering, and mathematics (STEM) nationwide. In 2024, the SERU survey examined multiple aspects of student life, including academic engagement, educational experiences, satisfaction, use of generative AI, academic and personal development, student well-being, plans and aspirations, cost of attendance, and background characteristics.

The survey was administered online and linked with administrative records containing demographic and academic information such as age, major (based on Classification of Instructional Programs (CIP) codes), and year of study. All undergraduate students enrolled at participating institutions were invited, resulting in an overall average response rate of approximately 21%, ranging from 11% to 37% across institutions. Informed consent was obtained electronically before participation. Overall, over 102,000 students participated in the survey. For this study, we used only completed surveys from students at the main university campuses, excluding partial responses and those from satellite campuses. This resulted in a final analytic sample of 95,513 undergraduate students across 20 institutions.

The survey results are broadly representative of the undergraduate populations at participating institutions across key academic and demographic characteristics (Table S1). The most notable discrepancy in representativeness involves gender, with female respondents overrepresented by eight percentage points and male respondents underrepresented by nine percentage points. This pattern aligns with typical response distributions observed in other student surveys (1). We conduct a sensitivity analysis by comparing results with and without gender weights. We find that the results do not change substantively (see Table S5) and report results for unweighted data in the main text.

1.2 Measures

The primary variables used in this study include both self-reported survey measures and administrative records:

- *Underrepresented Minority (URM) Status*: Determined from administrative data, coded as URM if a student identified as Hispanic, African American, American Indian, or Pacific Islander. Students identifying exclusively as White, Asian, or both White and Asian were categorized as non-URM.
- *Social Class*: Derived from self-reported responses to the question, “Which of the following best describes your social class when you were growing up?” Responses were aggregated into three categories: low-income (“Low-income or poor” and “Working class”), middle class (“Middle class”), and upper class (“Upper-middle class” and “Wealthy”).

- *Disability*: Self-reported binary measure based on responses indicating the presence or absence of conditions or disabilities that significantly affect academic or social experiences.
- *First-generation Status*: Based on self-reported parental education levels; students were categorized as first-generation only if they indicated that neither parent or guardian had attended any college.
- *Gender*: Obtained from administrative data, classified as male, female and other/unknown.
- *Major*: Categorized based on administrative CIP codes into broad disciplinary categories such as Physical Sciences, Engineering, Humanities, among others. STEM or non-STEM classification was created using the U.S. Department of Homeland Security (DHS) STEM Designated Degree Program List.
- *Use of Generative AI*: Based on responses to the question, “Generative artificial intelligence (AI) tools like ChatGPT are becoming more common in academic settings. During this academic year, how often have you used such tools?” Response options included: Never, Several times per year, Several times per month, Several times per week, and Daily.

These measures form the basis of analytical models designed to examine factors associated with generative AI usage among undergraduate students at research-intensive institutions.

2 Methods

2.1 Estimating GenAI-Assisted Cheating

2.1.1 List Experiment Design

In order to estimate GenAI-assisted cheating, we designed a list experiment (also known as item count technique or unmatched count technique). This methodology was initially developed to measure racial attitudes (2), and has since been used in a wide variety of settings where eliciting truthful answers to sensitive questions is important (3, 4).

Although direct anonymous surveys could, in principle, be used to estimate GenAI-assisted cheating, most large-scale student surveys avoid behavior-specific self-admission questions because respondents may be reluctant to disclose stigmatized conduct and may underreport even under anonymity. Consistent with this pattern, prior surveys of student AI use largely focus on perceptions or attitudes rather than direct admissions, and the few available direct benchmarks suggest low single-digit prevalence (e.g., 5), though they differ in national context, institutional type, and policy clarity.

Instead of asking a sensitive question directly such as “Have you ever submitted AI-generated content as your own work in class, knowing it may not be allowed?”—which survey respondents might be unwilling to answer truthfully—we randomly split the sample into two equal-sized groups (control and treatment) and asked the following question¹:

¹ Students’ understanding of permitted AI use may not align with course or institutional policy, so responses to this item may reflect uncertainty about what is allowed. Such uncertainty can introduce measurement error and likely yields conservative estimates of AI misuse. Students who submitted GenAI-assisted work that was prohibited but

“Review the following statements about using generative AI tools like ChatGPT and report how many — but not which — are true for you”

The control group was provided with the following three non-sensitive statements:

- 1. I have had a class with assigned readings on the ethics of generative AI.*
- 2. I have heard my classmates mention using AI tools like ChatGPT for their assignments.*
- 3. I have often explained to my classmates how to use AI tools like ChatGPT.*

The treatment group received the same three statements plus an additional sensitive statement:

- 4. I have submitted AI-generated content as my own work in class, knowing it may not be allowed.*

Those who answered “Never” to the question “During this academic year, how often have you used such tools?” were excluded from the list experiment, resulting in a sample size of 61,509 respondents.

2.1.2 Sample Construction and Randomization Balance

The randomization resulted in a balanced dataset, with no significant differences between the control and treatment groups across key demographic and academic characteristics (Table S2). P-values in Table S2 are unadjusted for multiple comparisons.

2.1.3 Design and Ceiling Effects Assessment

An important consideration when conducting list experiments is the potential for design effects. Design effects might appear when the probability of responding affirmatively to non-sensitive questions depends on the inclusion of the sensitive item. For the approach to be valid, this assumption can be loosened so that it is only required that the sum of positive responses to non-sensitive items doesn’t depend on the inclusion of the sensitive item (3, 4).

To test for design effects, we examined the conditional distribution of the sum of positive responses to non-sensitive items for treatment and control groups. Table S3 presents the estimated conditional probabilities. The Bonferroni-corrected p-value is 0.28, indicating that our dataset meets the no-design-effect assumption.

Another assumption in list randomization is that respondents provide a truthful response to the sensitive item (3, 4). Respondents who already endorse all non-sensitive items may lose anonymity when responding to the sensitive item and thus hide their truthful response, leading to ceiling-related underreporting. Table S4 reports the raw distribution of item counts by treatment status and academic discipline. Overall, 11% of respondents in the control group endorsed all three non-sensitive items, with most responses concentrated in the interior categories (44% endorsing one item and 35% endorsing two items), indicating substantial headroom for endorsement of the sensitive item.

believed it was permitted would be unlikely to endorse “knowing it may not be allowed,” whereas the reverse case is less likely.

Ceiling rates vary across disciplines from 7.5% to 17.2%. Although higher ceiling shares may increase the risk of attenuation through underreporting when respondents lack headroom (3), these values remain moderate and the response mass remains concentrated away from the ceiling.

As an illustrative upper-bound approximation, the maximum expected attenuation is bounded by the share of respondents at the control-list maximum. A rough approximation is (ceiling share) * (estimated prevalence). This implies a modest 1 pp effect overall (11% * ~9%) and roughly 1-2 pp in the highest-ceiling disciplines.

2.1.4 Estimation Strategy

We estimate GenAI-assisted cheating using several modeling strategies. Our primary method is a constrained maximum likelihood estimation (MLE) model for list experiments (3). This approach takes full advantage of the detailed response patterns for both control and sensitive items. Being “constrained” means that estimation is performed under restrictions that force the model’s predicted probabilities to remain consistent with the logical limits of the list experiment design. This allows us to extract more information from the full distribution of responses, leading to more precise and efficient estimates of the overall prevalence of affirmative responses.

Let $J = 3$ denote the number of non-sensitive (control) items. For respondent i , $T_i \in \{0,1\}$ indicates assignment to the control (0) or treatment (1) list; Y_i is the reported count of “yes” responses; X_i are observed covariates. Under the standard assumptions of no design effect and truthful response to the sensitive item, the potential outcomes satisfy:

$$Y_i(1) = Y_i(0) + Z_i^*,$$

where $Z_i^* \in \{0,1\}$ is the (latent) truthful answer to the sensitive item.

Unadjusted model (no covariates). Let $\pi = \Pr(Z_i^* = 1)$ denote the prevalence of the sensitive behavior. Following Blair and Imai (2012), we estimate π by constrained maximum likelihood using the full distribution of reported counts in the control and treatment groups, while modeling the distribution of the number of endorsed control items $Y_i(0)$ using the standard likelihood-based list experiment framework for J control items. Intuitively, the observed counts satisfy:

In the control group ($T_i = 0$), Y_i equals the number of endorsed control items $Y_i(0)$.

In the treatment group ($T_i = 1$), Y_i equals $Y_i(0)$ for respondents with $Z_i^* = 0$ and equals $Y_i(0) + 1$ for respondents with $Z_i^* = 1$.

Parameters are estimated by constrained maximum likelihood, ensuring predicted probabilities remain within the feasible range implied by the list experiment design.

Covariate-adjusted model. To allow the prevalence of the sensitive behavior to vary with student characteristics, we extend by allowing π to depend on observed covariates:

$$\pi_i = \text{logit}^{-1}(X_i \delta)$$

where X_i includes university affiliation, gender, age, year of study, STEM indicator, and frequency of AI use. Estimation proceeds using the same constrained maximum likelihood list experiment framework, with the prevalence parameter modeled as a function of X_i . Parameters

are estimated via constrained maximum likelihood, ensuring predicted probabilities remain consistent with the logical limits of the list experiment design.

The resulting estimates, presented in Table S5, are highly similar across both unadjusted and covariate-adjusted specifications, supporting the robustness of our findings.

Robustness Checks. In addition to the constrained MLE model, we also perform a series of robustness checks using alternative specifications:

- an unconstrained MLE model (does not impose the same logical restrictions as the constrained model),
- method-of-moments difference-in-means estimators (both weighted by gender and unweighted), and
- a cross-classified multilevel model with random intercepts for university and academic discipline.

For the regression-based models (the method-of-moments estimators), we include the same set of covariates and employ cluster-robust standard errors, clustered at the university level, to account for within-campus correlations.

Across all approaches, the resulting estimates presented in Table S5 largely overlap, confirming the robustness of our findings. Based on its efficiency, the constrained MLE model is used to generate discipline-specific estimates, as reported in the main text and Table S6.²

2.2 Estimating Disparities in GenAI Use

We use logistic regression to examine disparities in GenAI use across groups. Our focus is on the coefficients for gender, race/ethnicity, social class, first-generation status, and disability status. The model controls for the institution and year of study with university fixed effects and year-of-study indicators. Cluster-robust standard errors are used to account for within-university correlations. The model is specified as:

$$\text{Logit}(P(Y=1)) = \beta_0 + \beta_1(\text{Gender}) + \beta_2(\text{URM}) + \beta_3(\text{Social Class}) + \beta_4(\text{First-Generation}) + \beta_5(\text{Disability}) + \beta_6(\text{University}) + \beta_7(\text{Year})$$

Estimates are initially obtained as log odds and then exponentiated to produce odds ratios. We fitted the same model for each academic major category and for STEM versus non-STEM groups. The results are presented in Table S7.

² Because discipline-specific estimates rely on smaller samples than the overall estimate, they are naturally less precise. We therefore emphasize confidence intervals and interpret small cross-discipline differences cautiously.

Figure S1. Disparities in GenAI Use across Disciplines. Estimates are derived from separate university fixed-effects logistic regression models for each discipline, for STEM, non-STEM groups, and overall, with clustered standard errors, adjusting for gender, race/ethnicity, social class, first-generation status, disability status and year of study (N = 95,513). Adjusted odds ratios are shown for the groups named in each panel, relative to male students, non-URM students, students from upper-middle or wealthy backgrounds, and students without disabilities. Values below 1 indicate lower odds of regular GenAI use than in the reference group, holding other variables constant. Year of study and university are included as controls but not shown; first-generation status is omitted because estimates are not statistically significant. Black horizontal lines show 95% CIs.

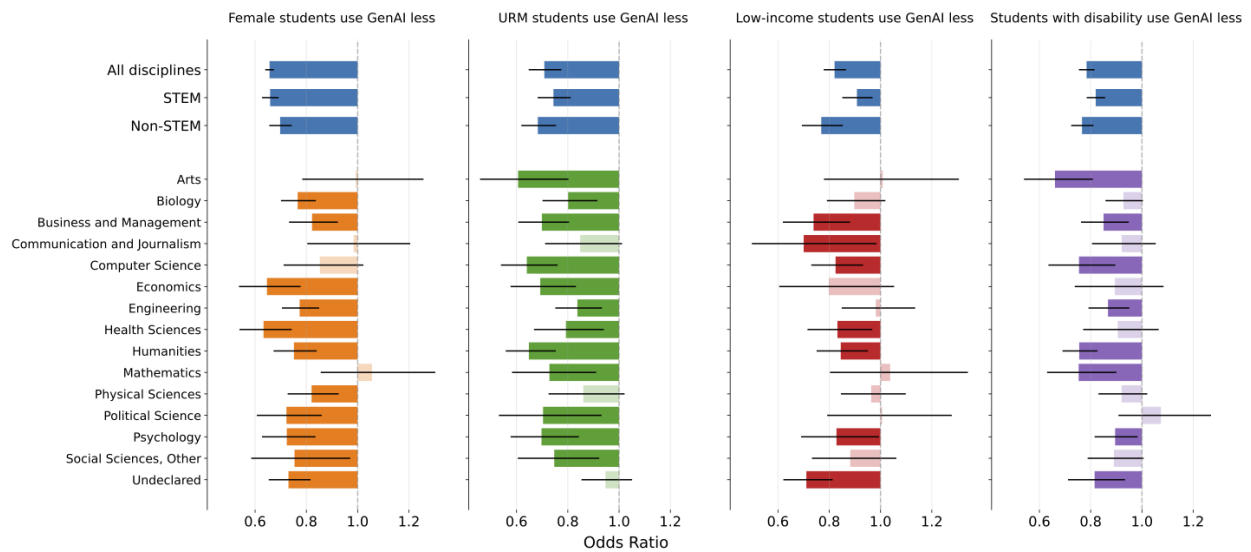


Table S1. Comparison of Study Population and Sample Characteristics by Subgroup

Characteristic	Respondents (%)	Population est (%)	Difference (perc. pt.)
Age			
18-19	38.6	37.5	1.1
20-21	43.1	43.7	-0.6
22 and above	18.3	18.8	-0.5
Gender			
Female	63.9	55.5	8.4
Male	34.3	43.6	-9.3
Other/Unknown	1.8	0.9	0.9
Race/Ethnicity			
African American	3.8	3.8	0.0
Asian	28.7	29	-0.3
Hispanic	24	23.2	0.8
Other	7.2	7.1	0.1
White	36.3	36.9	-0.6
Level of Study			
First-year	10.7	9.8	0.9
Sophomore	21.6	21.2	0.4
Junior	26.6	26.4	0.2
Senior	41.1	42.6	-1.5
Area of Study			
STEM	49.7	50.5	-0.8
Non-STEM	42.8	41.4	1.4
Undeclared	7.5	8.1	-0.6

Table S2. Balance Table for the List Randomization Experiment

	Control (N = 30,943)	Treatment (N = 30,566)	P-value
University			0.498
A	1536 (5.0)	1561 (5.1)	
B	1431 (4.6)	1363 (4.5)	
C	678 (2.2)	676 (2.2)	
D	950 (3.1)	924 (3.0)	
E	929 (3.0)	930 (3.0)	
F	866 (2.8)	856 (2.8)	
G	1352 (4.4)	1434 (4.7)	
H	541 (1.7)	494 (1.6)	
I	1551 (5.0)	1491 (4.9)	
J	2093 (6.8)	2130 (7.0)	
K	2017 (6.5)	2013 (6.6)	
L	3383 (10.9)	3346 (10.9)	
M	2121 (6.9)	2021 (6.6)	
N	2228 (7.2)	2189 (7.2)	
O	473 (1.5)	435 (1.4)	
P	1355 (4.4)	1345 (4.4)	
Q	2835 (9.2)	2885 (9.4)	
R	2121 (6.9)	1999 (6.5)	
S	1036 (3.3)	1099 (3.6)	
T	1447 (4.7)	1375 (4.5)	
Level of Study			0.776
First-year	3018 (9.8)	2929 (9.6)	
Sophomore	6525 (21.1)	6456 (21.1)	
Junior	8198 (26.5)	8041 (26.3)	
Senior	13194 (42.7)	13139 (43.0)	
Major			0.041
Arts	1076 (3.5)	951 (3.1)	
Biology	4086 (13.2)	3964 (13.0)	
Business and Management	2733 (8.8)	2773 (9.1)	
Communication and Journalism	899 (2.9)	904 (3.0)	
Computer Science	2147 (6.9)	2142 (7.0)	
Economics	974 (3.1)	941 (3.1)	
Engineering	3934 (12.7)	3904 (12.8)	
Health Sciences	1309 (4.2)	1350 (4.4)	
Humanities	2162 (7.0)	2033 (6.7)	
Mathematics	958 (3.1)	889 (2.9)	
Multidisciplinary Studies	2279 (7.4)	2312 (7.6)	
Other	323 (1.0)	324 (1.1)	
Physical Sciences	1525 (4.9)	1457 (4.8)	
Political Science	686 (2.2)	788 (2.6)	
Psychology	1869 (6.0)	1836 (6.0)	
Social Sciences, Other	1578 (5.1)	1653 (5.4)	

	Control (N = 30,943)	Treatment (N = 30,566)	P-value
Undeclared	2405 (7.8)	2345 (7.7)	
Area of Study			0.499
Non-STEM	12169 (39.3)	12134 (39.7)	
STEM	16412 (53.0)	16158 (52.9)	
Undeclared	2362 (7.6)	2274 (7.4)	
Gender			0.743
Female	19080 (61.7)	18806 (61.5)	
Male	11430 (36.9)	11352 (37.1)	
Other/unknown	430 (1.4)	407 (1.3)	
Age			0.640
18-19	11550 (37.3)	11404 (37.3)	
20-21	13939 (45.0)	13690 (44.8)	
22 and above	5454 (17.6)	5472 (17.9)	
URM (Yes)	7367 (26.1)	7308 (26.1)	0.963
Social Class			0.262
Low-income	8240 (30.5)	7963 (29.9)	
Middle class	10756 (39.8)	10754 (40.4)	
Upper class	8023 (29.7)	7920 (29.7)	
Disability (Yes)	8920 (32.1)	8722 (31.9)	0.565
First-Generation Status (Yes)	5047 (18.0)	4871 (17.6)	0.243
Frequency of GenAI Use			0.191
Daily	1942 (6.3)	2018 (6.6)	
Weekly	5877 (19.0)	5768 (18.9)	
Monthly	9802 (31.7)	9805 (32.1)	
Yearly	13322 (43.1)	12975 (42.4)	

Table S3. Test for List Experiment Design Effect

Outcome Category	Estimated Proportion	Standard Error
0 control-item yes responses ($Y_i(0)=0$, $Z_i=1$)	-0.0025	0.0023
1 control-item yes response ($Y_i(0)=1$, $Z_i=1$)	0.0409	0.0040
2 control-item yes responses ($Y_i(0)=2$, $Z_i=1$)	0.0240	0.0026
3 control-item yes responses ($Y_i(0)=3$, $Z_i=1$)	0.0149	0.0007
0 control-item yes responses ($Y_i(0)=0$, $Z_i=0$)	0.0938	0.0017
1 control-item yes response ($Y_i(0)=1$, $Z_i=0$)	0.4029	0.0033
2 control-item yes responses ($Y_i(0)=2$, $Z_i=0$)	0.3296	0.0034
3 control-item yes responses ($Y_i(0)=3$, $Z_i=0$)	0.0964	0.0019

Overall Test (Bonferroni-corrected p-value): 0.2785

Note: $Y_i(0)$ denotes the latent number of 'yes' responses to the control items, and Z_i denotes the latent binary response to the sensitive item (1 = yes, 0 = no). The table reports estimated population proportions of respondent types from the Blair and Imai (3) design-effects test, as implemented by the `ict.test()` function in the `list` R package. Inference is based on the joint Bonferroni-corrected p-value reported above.

Table S4. Response Distribution by Control/Treatment Group and Discipline

Category	Group	0 items (%)	1 item (%)	2 items (%)	3 items (%)	4 items (%)	Group N	N
Overall	Control	9.1	44.4	35.4	11.1	0.0	30,943	61,509
	Treatment	9.4	40.0	37.1	12.0	1.5	30,566	
Arts	Control	10.6	42.0	36.3	11.1	0.0	1,076	2,027
	Treatment	9.8	39.9	36.6	12.2	1.6	951	
Biology	Control	10.5	47.3	33.9	8.3	0.0	4,086	8,050
	Treatment	11.0	45.2	33.9	9.2	0.7	3,964	
Business and Management	Control	6.5	37.4	39.4	16.7	0.0	2,733	5,506
	Treatment	7.3	33.0	41.0	15.9	2.8	2,773	
Communication and Journalism	Control	8.0	32.4	42.4	17.2	0.0	899	1,803
	Treatment	5.1	29.3	43.4	18.8	3.4	904	
Computer Science	Control	6.6	37.4	39.8	16.2	0.0	2,147	4,289
	Treatment	7.3	34.1	38.9	17.8	1.9	2,142	
Economics	Control	10.5	46.7	33.6	9.2	0.0	974	1,915
	Treatment	8.8	38.3	38.4	12.2	2.3	941	
Engineering	Control	9.3	46.3	34.7	9.7	0.0	3,934	7,838
	Treatment	9.2	40.8	37.5	11.4	1.1	3,904	
Health Sciences	Control	10.8	48.9	32.2	8.1	0.0	1,309	2,659
	Treatment	10.3	43.9	35.1	9.6	1.0	1,350	
Humanities	Control	10.4	46.1	33.9	9.7	0.0	2,162	4,195
	Treatment	12.3	42.6	33.1	10.7	1.2	2,033	
Mathematics	Control	10.3	49.7	29.3	10.6	0.0	958	1,847
	Treatment	10.7	41.4	36.2	10.2	1.5	889	
Multidisciplinary Studies	Control	9.2	41.6	36.4	12.8	0.0	2,279	4,591
	Treatment	8.7	38.2	38.7	13.1	1.3	2,312	
Other	Control	10.2	48.3	31.6	9.9	0.0	323	647
	Treatment	12.3	39.8	36.1	9.9	1.9	324	
Physical Sciences	Control	9.2	53.7	29.5	7.5	0.0	1,525	2,982
	Treatment	10.0	44.5	35.1	9.3	1.1	1,457	
Political Science	Control	8.2	46.1	34.3	11.5	0.0	686	1,474
	Treatment	9.5	41.6	34.8	12.8	1.3	788	
Psychology	Control	10.4	45.3	34.7	9.6	0.0	1,869	3,705
	Treatment	9.7	44.1	34.4	10.2	1.6	1,836	
Social Sciences, Other	Control	8.9	44.7	34.9	11.5	0.0	1,578	3,231
	Treatment	9.8	40.0	37.7	10.8	1.6	1,653	
Undeclared	Control	7.6	43.9	37.8	10.7	0.0	2,405	4,750
	Treatment	8.6	39.1	39.5	11.6	1.2	2,345	

Table S5. Summary of Estimates for GenAI-Assisted Cheating Prevalence

Model	Estimate	N
Constrained MLE, no covariates	0.09 (0.004)	61,509
Constrained MLE, with covariates	0.08 (0.004)	61,500
Unconstrained MLE, no covariates	0.07 (0.008)	61,509
Method of moments, unweighted, no covariates	0.08 (0.007)	61,509
Method of moments, unweighted, with covariates	0.07 (0.005)	61,500
Method of moments, weighted, no covariates	0.08 (0.007)	61,509
Method of moments, weighted, with covariates	0.08 (0.006)	61,500
Multilevel (cross-classified), no covariates	0.08 (0.006)	61,509
Multilevel (cross-classified), with covariates	0.07 (0.007)	61,500

Note: Standard errors are in parentheses. Sample sizes (N) vary slightly across models due to missing values in covariates, affecting fewer than 0.02% of observations.

Table S6. Regular GenAI Use (monthly or more) and GenAI-Assisted Cheating Among GenAI Users by Academic Discipline

Category	Regular Gen-AI Use, %	Regular Gen-AI Use, CI	Gen-AI Assisted Cheating, %	Gen-AI Assisted Cheating, CI
All disciplines	37.1	[36.8, 37.4]	9.3	[8.5, 10.0]
STEM	41.6	[41.2, 42.1]	8.3	[7.4, 9.4]
Non-STEM	31.8	[31.3, 32.2]	10.8	[9.5, 12.0]
Computer Science	61.5	[60.2, 62.8]	10.0	[7.1, 13.0]
Mathematics	52.4	[50.5, 54.4]	10.1	[5.5, 15.7]
Business and Management	50.8	[49.7, 52.0]	12.8	[10.3, 15.7]
Economics	48.6	[46.7, 50.6]	17.0	[10.9, 23.6]
Engineering	46.0	[45.1, 46.9]	7.5	[5.5, 9.6]
Undeclared	38.0	[36.9, 39.1]	7.0	[4.6, 9.6]
Communication and Journalism	36.8	[34.9, 38.6]	16.5	[11.6, 22.1]
Physical Sciences	34.0	[32.7, 35.4]	8.5	[4.9, 12.5]
Biology	33.3	[32.5, 34.1]	4.8	[3.1, 6.3]
Health Sciences	29.8	[28.5, 31.2]	7.7	[4.2, 11.9]
Psychology	29.0	[27.9, 30.2]	9.7	[6.7, 13.0]
Political Science	28.3	[26.6, 30.1]	8.3	[3.9, 13.4]
Humanities	26.4	[25.5, 27.4]	8.1	[5.2, 11.0]
Social Sciences, Other	25.8	[24.7, 26.9]	9.1	[6.2, 12.4]
Arts	23.6	[22.3, 24.9]	10.0	[5.6, 14.8]

Table S7. Regression Results for Disparities in GenAI Use across Disciplines

Discipline	Variable	Odds Ratio	95% CI	p-value
Humanities	Gender (Female)	0.75	[0.67, 0.84]	0.000
Humanities	URM (Yes)	0.65	[0.56, 0.75]	0.000
Humanities	Social class (Low-income)	0.84	[0.75, 0.95]	0.005
Humanities	First-generation Status (Yes)	1.10	[0.96, 1.25]	0.165
Humanities	Disability (Yes)	0.76	[0.69, 0.83]	0.000
Physical Sciences	Gender (Female)	0.82	[0.73, 0.93]	0.001
Physical Sciences	URM (Yes)	0.86	[0.72, 1.02]	0.086
Physical Sciences	Social class (Low-income)	0.96	[0.85, 1.1]	0.578
Physical Sciences	First-generation Status (Yes)	1.01	[0.85, 1.18]	0.946
Physical Sciences	Disability (Yes)	0.92	[0.83, 1.02]	0.117
Arts	Gender (Female)	0.99	[0.78, 1.26]	0.951
Arts	URM (Yes)	0.61	[0.46, 0.8]	0.000
Arts	Social class (Low-income)	1.01	[0.78, 1.31]	0.948
Arts	First-generation Status (Yes)	1.30	[0.95, 1.77]	0.098
Arts	Disability (Yes)	0.66	[0.54, 0.81]	0.000
Health Sciences	Gender (Female)	0.63	[0.54, 0.74]	0.000
Health Sciences	URM (Yes)	0.79	[0.67, 0.94]	0.008
Health Sciences	Social class (Low-income)	0.83	[0.72, 0.97]	0.017
Health Sciences	First-generation Status (Yes)	0.91	[0.73, 1.15]	0.441
Health Sciences	Disability (Yes)	0.91	[0.77, 1.07]	0.232
Communication and Journalism	Gender (Female)	0.98	[0.8, 1.21]	0.876
Communication and Journalism	URM (Yes)	0.85	[0.71, 1.01]	0.067
Communication and Journalism	Social class (Low-income)	0.70	[0.5, 0.98]	0.040
Communication and Journalism	First-generation Status (Yes)	1.10	[0.86, 1.41]	0.433
Communication and Journalism	Disability (Yes)	0.92	[0.81, 1.05]	0.232
Computer Science	Gender (Female)	0.85	[0.71, 1.02]	0.085
Computer Science	URM (Yes)	0.64	[0.54, 0.76]	0.000
Computer Science	Social class (Low-income)	0.82	[0.73, 0.93]	0.002
Computer Science	First-generation Status (Yes)	1.15	[0.93, 1.42]	0.192
Computer Science	Disability (Yes)	0.75	[0.64, 0.9]	0.001
Political Science	Gender (Female)	0.72	[0.61, 0.86]	0.000
Political Science	URM (Yes)	0.70	[0.53, 0.93]	0.014
Political Science	Social class (Low-income)	1.01	[0.79, 1.28]	0.960
Political Science	First-generation Status (Yes)	0.97	[0.7, 1.36]	0.881
Political Science	Disability (Yes)	1.07	[0.91, 1.27]	0.405
Social Sciences, Other	Gender (Female)	0.75	[0.59, 0.97]	0.029
Social Sciences, Other	URM (Yes)	0.75	[0.61, 0.92]	0.007
Social Sciences, Other	Social class (Low-income)	0.88	[0.73, 1.06]	0.185
Social Sciences, Other	First-generation Status (Yes)	1.02	[0.86, 1.21]	0.846
Social Sciences, Other	Disability (Yes)	0.89	[0.79, 1.01]	0.063
Mathematics	Gender (Female)	1.06	[0.86, 1.3]	0.611
Mathematics	URM (Yes)	0.73	[0.58, 0.91]	0.005
Mathematics	Social class (Low-income)	1.04	[0.8, 1.34]	0.778
Mathematics	First-generation Status (Yes)	0.87	[0.69, 1.08]	0.207
Mathematics	Disability (Yes)	0.75	[0.63, 0.9]	0.002
Engineering	Gender (Female)	0.77	[0.7, 0.85]	0.000
Engineering	URM (Yes)	0.84	[0.75, 0.93]	0.001
Engineering	Social class (Low-income)	0.98	[0.85, 1.14]	0.800
Engineering	First-generation Status (Yes)	0.91	[0.79, 1.06]	0.230
Engineering	Disability (Yes)	0.87	[0.79, 0.95]	0.002
Economics	Gender (Female)	0.65	[0.54, 0.78]	0.000
Economics	URM (Yes)	0.69	[0.58, 0.83]	0.000
Economics	Social class (Low-income)	0.80	[0.6, 1.05]	0.111
Economics	First-generation Status (Yes)	1.01	[0.7, 1.46]	0.964

Discipline	Variable	Odds Ratio	95% CI	p-value
Economics	Disability (Yes)	0.89	[0.74, 1.08]	0.256
Business and Management	Gender (Female)	0.82	[0.73, 0.92]	0.001
Business and Management	URM (Yes)	0.70	[0.61, 0.8]	0.000
Business and Management	Social class (Low-income)	0.74	[0.62, 0.88]	0.001
Business and Management	First-generation Status (Yes)	0.95	[0.8, 1.14]	0.602
Business and Management	Disability (Yes)	0.85	[0.76, 0.95]	0.004
Psychology	Gender (Female)	0.72	[0.63, 0.84]	0.000
Psychology	URM (Yes)	0.70	[0.58, 0.84]	0.000
Psychology	Social class (Low-income)	0.83	[0.69, 0.99]	0.043
Psychology	First-generation Status (Yes)	0.94	[0.82, 1.09]	0.404
Psychology	Disability (Yes)	0.90	[0.82, 0.98]	0.020
Biology	Gender (Female)	0.77	[0.7, 0.84]	0.000
Biology	URM (Yes)	0.80	[0.7, 0.92]	0.001
Biology	Social class (Low-income)	0.90	[0.79, 1.02]	0.095
Biology	First-generation Status (Yes)	0.99	[0.86, 1.13]	0.870
Biology	Disability (Yes)	0.93	[0.86, 1]	0.062
Undeclared	Gender (Female)	0.73	[0.65, 0.82]	0.000
Undeclared	URM (Yes)	0.95	[0.85, 1.05]	0.305
Undeclared	Social class (Low-income)	0.71	[0.62, 0.81]	0.000
Undeclared	First-generation Status (Yes)	1.01	[0.87, 1.17]	0.892
Undeclared	Disability (Yes)	0.82	[0.71, 0.93]	0.003
STEM	Gender (Female)	0.66	[0.63, 0.69]	0.000
STEM	URM (Yes)	0.74	[0.68, 0.81]	0.000
STEM	Social class (Low-income)	0.91	[0.85, 0.97]	0.004
STEM	First-generation Status (Yes)	0.98	[0.92, 1.05]	0.610
STEM	Disability (Yes)	0.82	[0.78, 0.86]	0.000
Non-STEM	Gender (Female)	0.70	[0.66, 0.74]	0.000
Non-STEM	URM (Yes)	0.68	[0.62, 0.75]	0.000
Non-STEM	Social class (Low-income)	0.77	[0.69, 0.85]	0.000
Non-STEM	First-generation Status (Yes)	1.07	[1.01, 1.14]	0.025
Non-STEM	Disability (Yes)	0.77	[0.72, 0.81]	0.000
Overall	Gender (Female)	0.66	[0.64, 0.67]	0.000
Overall	URM (Yes)	0.71	[0.65, 0.77]	0.000
Overall	Social class (Low-income)	0.82	[0.78, 0.87]	0.000
Overall	First-generation Status (Yes)	1.02	[0.97, 1.07]	0.544
Overall	Disability (Yes)	0.78	[0.75, 0.82]	0.000

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